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Urine glyphosate level as a quantitative biomarker of oral exposure

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ABSTRACT

Background: Since the classification of glyphosate as a Group 2A substance "probably carcinogenic to humans" by the IARC in 2015, human health concerns have been raised regarding the exposure of operators, bystanders, and consumers. Urine measurement studies have been conducted, but since toxicokinetic data on glyphosate in humans is lacking, a meaningful interpretation of this data regarding exposure is not possible.
Objective: This study aims to determine the fraction of glyphosate and AMPA excretion in urine after consuming ordinary food with glyphosate residue, to estimate dietary glyphosate intake.
Methods: Twelve participants consumed a test meal with a known amount of glyphosate residue and a small amount of AMPA. Urinary excretion was examined for the next 48 h.
Results: Only 1% of the glyphosate dose was excreted in urine. The urinary data indicated the elimination half-life was 9 h. For AMPA, 23% of the dose was excreted in urine, assuming that no metabolism of glyphosate to AMPA occurred. If all of the excreted AMPA was a glyphosate metabolite, this corresponds to 0.3% of the glyphosate dose on a molar basis.
Conclusion: This study provides a basis for estimating oral glyphosate intake using urinary biomonitoring data.

1. Introduction

N-(phosphonomethyl)-glycine (glyphosate, CAS RN® 1071-83-6) is a systemic herbicide that competitively inhibits the enzyme 5-enolpyruvylshikimate-3-phosphate synthase, blocking plant biosynthesis of aromatic amino acids.

The main environmental biodegradation product of glyphosate is aminomethylphosphonic acid (AMPA, CAS RN® 1066-51-9). Glyphosate is poorly metabolised in animals and plants, and AMPA is the main metabolite (EFSA, 2015).

Human health concerns have been raised regarding the exposure of operators, bystanders, and residents to glyphosate-based pesticides during spraying and consumer exposure to glyphosate residues in food crops (Myers et al., 2016). The International Agency for Research on Cancer (IARC, 2015) recently concluded that "Glyphosate is probably carcinogenic to humans (Group 2A)". According to the European Chemicals Agency's (ECHA) Committee for Risk Assessment (RAC), glyphosate does not meet the classification criteria as a carcinogen, a mutagen, or for reproductive toxicity (ECHA, 2017). The reasons for the diverging conclusions regarding glyphosate's carcinogenic hazard were recently reviewed (Portier et al., 2016; Tarazona et al., 2017). The European Food Safety Authority (EFSA) recommended an acceptable daily intake (ADI) and an acute reference dose (ARfD) of 0.5 mg kg^{-1} body weight (bw) per day (EFSA, 2015). The Joint FAO/WHO Meeting on Pesticide Residues (JMPR) recommended an ADI of 0–1 mg kg^{-1} bw per day and considered an ARfD unnecessary due to glyphosate's low acute toxicity (FAO/WHO, 2017). Both bodies concluded that glyphosate and AMPA have similar toxic potency and the maximum residue levels set for glyphosate and expected consumer exposure to food crop residues are safe. Swiss consumer exposure to glyphosate was demonstrated in a limited survey (RTS, 2015). In approximately 40% of 40 participants, glyphosate was detected in the urine at levels of 0.1–1.5 ng mL^{-1}. Analyses in Germany confirmed low levels of glyphosate and its degradation product AMPA in consumers' urine (Conrad et al., 2017). In approximately 30–50% of the urine samples, glyphosate and AMPA were above the LOQ of 0.1 ng mL^{-1}. The median levels of both glyphosate and AMPA were below the LOQ or up to 0.2 ng mL^{-1} and the maximum values were 2.8 and 1.9 ng mL^{-1}, respectively. The highest quantification rates and highest values were reported in 2012 and 2013. There was a moderate correlation between glyphosate values and AMPA. Based on this urinary data, glyphosate exposure in Switzerland and Germany appear to be similar.

An analysis of glyphosate and AMPA in food commodities in Switzerland, identified cereals and pulses (e.g., beans, lentils, chickpeas) as the main contributors for consumers' exposure (Zoller et al., 2018). Although only approximately 20% of an orally administered

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Abbreviations

ADI	acceptable daily intake
AMPA	aminomethylphosphonic acid
ARfD	acute reference dose
bw	body weight
E_{max}	maximum excretion rate
F	female
LOD	limit of detection
LOQ	limit of quantification
M	male
T_{max}	time to excretion peak

glyphosate dose is absorbed in laboratory animals (EFSA, 2015), urine levels of glyphosate are considered to be good exposure markers because glyphosate does not bioaccumulate and is poorly metabolised and exclusively to AMPA. Based on rat absorption rates, a preliminary dietary risk assessment of glyphosate and AMPA levels was conducted and no unacceptable health risk was identified (Zoller et al., 2018). The validity of this dietary risk assessment was limited by using rat oral absorption rates as a proxy for the human oral absorption rate. However, animal absorption rates are not reliable to predict human oral absorption (Musther et al., 2014) and may compromise risk assessments based on oral exposure. Using the animal oral absorption rate is especially problematic if it is low, as in the case of glyphosate. If the unknown human absorption rate is significantly higher, the risk to consumers is underestimated. If urinary excretion represents the systemically available dose, this uncertainty can be reduced by comparing human urinary levels to a known intake level.

The scope of the present study was to correlate urinary glyphosate and AMPA excretion with a known oral glyphosate and AMPA dose in a natural matrix. This allows the use of urinary glyphosate and AMPA levels as reliable quantitative biomarkers of oral exposure to obtain a more refined human exposure assessment.

2. Materials and methods

2.1. Study design and protocols

The four-day trial was designed as follows: after a two-day wash-out period, glyphosate was administered via a test meal followed by a two-day urine sampling period. During the trial, the participants' glyphosate intake was kept as low as possible to enable the observation of the excretion of the administered glyphosate.

The participants received two lists (see Supplementary Information S1), one that has foods that are assumed to have low glyphosate residue levels, which were recommended for consumption and the other with products to avoid because of possible glyphosate contamination. A few days later, shortly before the trial started, the participants were individually instructed on the study protocol and the recommended foods were discussed in detail. The participants were instructed to eat according to the food lists during the four-day trial. They were also instructed to eat as similar and monotonous meals as possible over the trial period and to keep a detailed log of their food and beverage consumption. During the two-day urine sample collection period, each micturition was logged and collected in a separate 1000 mL polypropylene wide-mouth bottle (Semadeni, Ostermundigen, Switzerland), with the time and date recorded on the label, and stored in a refrigerator. The samples were transferred to the laboratory within a few hours or at a maximum, within 24 h. In the laboratory, the volume and pH of each sample was measured. Several 5 mL aliquots were taken and either used for immediate analysis or stored in the freezer at −20 °C until analysis.

The test meal was eaten in the morning of a workday at the laboratory, so the most important excretion period after the meal was during the day. The participants were instructed to log the exact time of micturition of the first morning urine at home, but not to collect it. They were told to drink a little and eat little to nothing, to ensure they would be able to consume the test portion of approximately 150 g of falafel. Shortly before the test meal, the first urine was collected. Most of the participants also drank water while eating the test meal.

The participants were instructed to urinate regularly once every 1.5 h after the test meal to obtain enough data points during the first 6 h after consumption of the glyphosate dose. They were instructed to drink enough to allow sufficient urination during the day but to avoid excessive consumption of water that could hamper the glyphosate analysis with too strong dilution. After the initial 6-h period, the participants were free to urinate as needed. Each urine sample was collected in a separate bottle until the first morning urination two days after the test meal.

The study protocol was approved by the Ethics Committee of Bern (BASEC ID 2018–00110). Participation was voluntary and informed consent was obtained from all of the participants.

2.2. Study participants

Office employees working near the laboratory were recruited as volunteers. All of the participants received a voucher worth CHF 200 for a local retailer. A short walking distance to the laboratory was defined as the selection criterion for ease of handling of the urine samples. Twelve healthy adults, six males (M) and six females (F) participated.

2.3. Test meal

The test meal was a homemade falafel dish. Each participant was instructed to eat approximately 150 g of the falafel, corresponding to an intake of 196.8 µg of glyphosate and 1.67 µg of AMPA. The falafel's ingredients met all the legal requirements. The food used was not spiked with glyphosate or AMPA. Both substances were present as residues in the gram flour (chickpea flour) used (see Supplementary Information S2). The falafel dish was prepared ready for consumption. All portions of the dish were frozen. For the trial, a portion of the dish was thawed and reheated in a combi steamer. The ingested glyphosate and AMPA dose were calculated before reheating. The falafel's homogeneity and stability were checked during the reheating process of the meal preparation (see Supplementary Information S3).

2.4. Analysis methods

The glyphosate and AMPA in the gram flour and falafel were analysed using a previously established method (Zoller et al., 2018). To analyse the glyphosate and AMPA in the urine samples, the same method was used with minor modifications (see Supplementary Information S4). The limit of quantification (LOQ) was defined as a signal to noise (s/n) ratio of 10:1 for the quantifier transition and the limit of detection (LOD) was 3:1. The LOQ for the glyphosate and AMPA was 0.05 and 0.1 ng mL^{-1}, respectively, and the LOD was 0.02 and 0.04 ng mL^{-1}, respectively. Values between the LOD and LOQ were also used for calculations. For values below the LOD, the following procedure was applied. If there was an integrable signal, this was used for calculation. Otherwise, the value was set to 0.01 ng mL^{-1}. Creatinine was analysed at an external clinical laboratory using the Jaffé Gen.2 method (cobas, Roche Diagnostics, Mannheim, Germany).

3. Results

All of the participants followed the study protocol and successfully completed the experiment. The experimental design showed to be suitable for obtaining sufficiently low background levels of glyphosate, as the concentration in the urine at time zero (before consumption of the test meal) was below 0.1 ng mL^{-1} for all participants. The experimental data is summarised in Table 1 and the raw data is available in Excel file S5. The administered dose range per participant was 125–247 μg for glyphosate and 1.06–2.08 μg for AMPA, which corresponds to the consumption of 95–188 g of falafel. The range of the applied dose in relation to body weight was 2010–4360 ng kg^{-1} for glyphosate and 17.0–39.2 ng kg^{-1} for AMPA. The highest urinary glyphosate concentration was 5.6 ng mL^{-1}, which was measured in the urine sample at the time interval of 9.92–11.50 h from participant 15M. The highest excretion rate of glyphosate was 250.6 ng h^{-1}, measured during the time interval of 8.25–9.92 h from participant 15M. The lowest and highest measured urine output rates were 13 mL h^{-1} and 569 mL h^{-1}, respectively. The mean urine sampling duration was 45.3 h within a range of 37–51 h.

The range of total urinary excretion in relation to the administered dose was 0.57–1.68% for glyphosate and 9.8–32.6% for AMPA when referring to the AMPA dose, or 0.087–0.276% when referring to the glyphosate dose. These results are based on the measured raw data without any data fitting. For participants 17F and 18F, one and two samples (micturitions), respectively, were missing. To estimate the magnitude of the error on glyphosate excretion, a simple linear interpolation with the adjacent excretion values was conducted to generate the missing data. This correction showed that the excretions were underestimated by approximately 3% and 9% for 17F and 18F, respectively. As this error was marginal, the data was used without correction.

The results showed no indication that the glyphosate and AMPA excretion rates increased due to higher urine output.

The participants' cumulative glyphosate and AMPA excretion are depicted in Fig. 1a and b. Most of the glyphosate and AMPA excretion was completed within 10–20 h after ingestion. After 45 h, the excretion of all of the participants reached a plateau. The cumulative excretion of glyphosate and AMPA expressed as a percentage of the dose was not significantly different in the male and female participants, according to the two-sample *t*-test (unequal variances). The p-values of glyphosate and AMPA were 0.31 and 0.14, respectively.

To calculate the toxicokinetic parameters, a one-compartment absorption model was applied. This model, shown in Fig. 2, assumes that both the oral absorption (Ka) and elimination rate (Ke) of glyphosate, followed a first-order kinetic (Hacker et al., 2009). The measured excretion values were fitted against the equation using the non-linear least squares method. This simple model fitted the experimental data well for all of the participants (see Fig. 3) and the excretion rates were accurately assessed. The calculated toxicokinetic parameters are summarised in Table 1. A very consistent elimination half-life value was obtained. The half-life was approximately 9 h for both the male and female participants, but the variation differed. For the male participants, the observed half-life range was between 6.34 and 9.96 h whereas it varied from 5.21 to 14.49 h for the female participants. For almost all of the participants, the time to excretion peak (T_{max}) was reached within approximately 5 h. Two male participants had a shorter and a longer T_{max} of 2.99 h and 9.14 h. The maximum glyphosate excretion rate (E_{max}) varied from 45.1 to 212.45 ng h^{-1} with a median value of 94.85 ng h^{-1}. The E_{max} was slightly lower in the females than the males, but the females also received a lower dose on average.

Due to the very low AMPA concentration in the urine (see Fig. 3), it was not possible to calculate the toxicokinetic parameters of each individual participant. The combined data showed the absorption and elimination of AMPA, which followed similar kinetics as that of glyphosate (see Fig. 4a). The average calculated parameters are provided in Table 2.

4. Discussion

4.1. Glyphosate

The literature on glyphosate's toxicokinetics is surprisingly scant, especially in view of the public interest of the possible risk associated with oral consumption of pesticide residues due to consumption of food treated with glyphosate-based pesticides. Specifically, there are no publications discussing urinary glyphosate levels in relation to defined oral exposure. Only one paper explored glyphosate's half-life in human urine samples from amenity horticulture workers using glyphosate-based pesticides (Connolly et al., 2019). A symposium abstract also reported preliminary results of a kinetic study (Faniband et al., 2017). Glyphosate was administered at a dose level of 25% of the ADI (0.125 mg kg^{-1} bw) to one male and female subject and their urinary excretion was investigated for 100 h.

To the best of the author's knowledge, the present paper is the first to investigate glyphosate's urinary excretion characteristics in humans

Table 1

Doses of glyphosate given to individual participants and the observed excretion and excretion parameters. Excreted dose corresponds to the effective measured amount without any data fitting. C_{max} are the effectively measured maximum concentrations in relation to the creatinine concentration. T_{max} and E_{max} are the time to excretion peak and the maximum excretion rate, respectively. $T_{1/2}$ is the elimination half-life. These three parameters were calculated using the model shown in Fig. 2. Codes with M represent male participants and codes with F represent female participants.

Participant code	Glyphosate							AMPA	
	Urine sampling time period (h)	Ingested Dose (ng)	Excreted Dose (%)	C_{max} (ng/g creatinine)	T_{max} (h)	E_{max} (ng/h)	$T_{1/2}$ (h)	Ingested Dose (ng)	C_{max} (ng/g creatinine)
11M	44.42	188928	0.74	1876	5.08	90.6	7.46	1598	542
12M	46.08	216480	1.28	1865	5.71	140.2	9.96	1832	534
13M	46.50	205984	0.64	1471	5.60	72.3	9.16	1743	422
14M	44.77	173184	0.98	2120	4.19	92.8	9.23	1465	287
15M	47.00	246656	1.68	4256	9.14	212.5	6.34	2087	326
22M	36.92	233536	0.99	2673	2.99	165.6	8.09	2076	497
16F	44.25	245344	0.57	2105	4.40	69.2	14.49	1055	575
17F	45.42	124640	1.30	2825	4.97	114.9	5.21	1277	448
18F	45.92	150880	0.65	1084	4.76	45.1	10.05	1376	524
19F	45.58	162688	0.87	3006	4.50	109.6	5.59	1343	524
20F	46.33	158752	0.76	1484	4.57	56.4	13.18	1665	528
21F	50.58	196800	0.95	2157	4.99	96.9	9.21	1976	524
Median all	45.75	192864	0.91	2112	4.87	94.9	9.19	1632	524
Median M	45.43	211232	0.98	1998	5.34	116.5	8.63	1788	460
Median F	45.75	160720	0.82	2131	4.67	83.1	9.63	1360	524

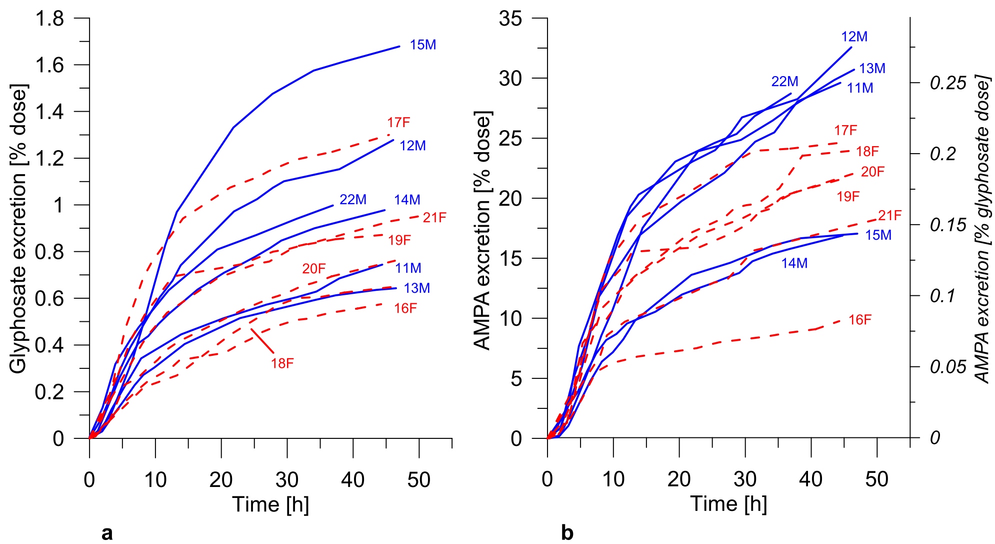

Fig. 1. Cumulative excretion of glyphosate (a) and AMPA (b) by male [blue, solid line] and female [red, dashed line] subjects expressed in percent of the oral dose.

$$Excretion = F\left(\frac{k_a}{k_a - k_e}\right) * (e^{-k_a * t} - e^{-k_e * t})$$

Fig. 2. One-compartment absorption model. F is a constant related to the bioavailability of the oral dose, k_a and k_e, the first-order absorption and elimination rate constant, respectively.

after oral consumption of a defined dose. In this study, to create a scenario as similar as possible to the consumer situation, glyphosate was administered as a real residue present in a foodstuff. Each urine sample was collected individually, so the precise time course could be evaluated. This data indicates that human urinary levels are a reliable indicator of oral dietary exposure to glyphosate residues. As the median urinary excretion was 0.91% (mean = 0.95%) of the administered dose, it is reasonable to assume that the urine levels indicated approximately 1% of dietary glyphosate exposure. Common dietary risk assessments compare the dose ingested — and not the dose systemically available after absorption — with a reference dose such as ADI or ARfD. Using nominal dietary exposure (glyphosate level in food multiplied by its consumption) instead of the systemically available dose, the procedure implicitly assumes 100% oral absorption. For most compounds, probably including glyphosate, this is an overestimation of the absorption and hence provides a certain margin of conservatism.

The observed median urinary excretion of 0.91% was 22 times lower than reported in animal studies, which showed a 20% excretion (EFSA, 2015). However, in animal studies, the dose was administered via gavage and not by feeding; thus, absorption might be lower. The range of the urinary excretion in different studies was from 11% to 43% (EFSA, 2014; WHO, 1994). Only one study administered glyphosate by diet (WHO, 1994). In this 14-day oral study in rats, the observed total excretion in urine was ≤10%. In the human study of Faniband et al. (2017), the dose was 30–60 times higher than in our study and the urinary excretion was similar. The total dose recovered as glyphosate in urine was approximately 1% and 0.4% in the female and male volunteers, respectively.

Dietary glyphosate is likely to be very poorly absorbed in humans and lower than in rats exposed by gavage. As the blood concentration was not measured in our study, this precluded any quantitative comparison of the toxicokinetic characteristics between rats and humans. However, the low urinary excretion level identified in our study (approximately 1% of dietary exposure) suggests that intake estimations calculated from human urine data systematically underestimated

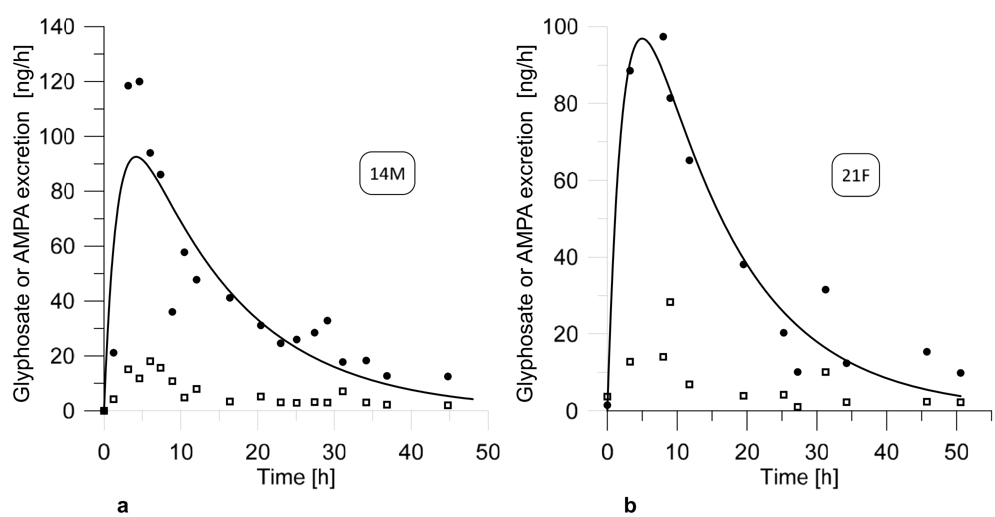

Fig. 3. Glyphosate (●) and AMPA (◘) excretion rate as a function of time for participants 14M (a) and 21F (b). Glyphosate data points are best fitted (solid line) with a one-compartment model, first-order absorption and elimination kinetics.

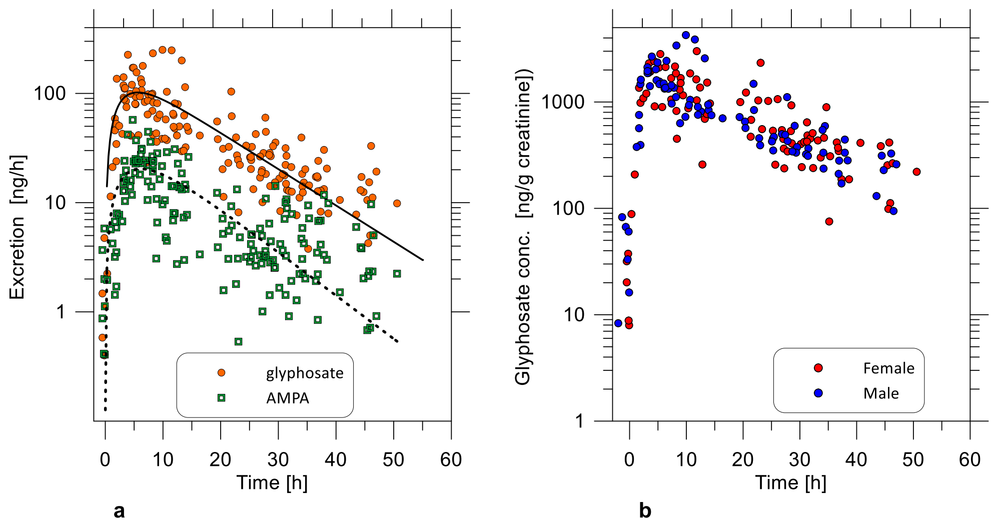

Fig. 4. a: observed excretion rates of glyphosate (◯) and AMPA (□); values are fitted (lines) with the model shown in Fig. 2 b: creatinine adjusted concentration of glyphosate for the male (blue) and female (red) subjects.

Table 2
Kinetics parameters for glyphosate and AMPA excretion calculated using the participants' data sets. F is a constant related to the bioavailability of the oral dose, K_a and K_e, the first-order absorption and elimination rate constant, respectively. T_{max} and E_{max} are the time to excretion peak and the maximum excretion rate, respectively. $T_{1/2}$ is the elimination half-life.

	Glyphosate	AMPA
F (ng/h)	156.89	35.67
K_a (h^{-1})	0.34	0.27
K_e (h^{-1})	0.08	0.09
T_{max} (h)	5.62	6.13
E_{max} (ng/h)	102.01	20.39
$T_{1/2}$ (h^{-1})	9.05	7.60

exposure (Conrad et al., 2017; Niemann et al., 2015).

The median urinary excretion of 0.91% of the dose has some uncertainties. The mean dose per person of 192 µg of glyphosate was quite low but comparable to assumed regular intake of glyphosate. During the 4-day experiment, the participants were free to consume recommended foods that were likely to contain only very low amounts of glyphosate. Their food consumption was not fully controlled, only recorded. Since the foods they ingested were not tested for glyphosate, there is a potential of consuming additional small amounts of glyphosate that could cause the underestimation of the dietary exposure. However, all of the participants had very low glyphosate concentrations after the wash-out period, indicating that the diet regimen was adequate. Additionally, the cumulative excretion curves (Fig. 1a) do not provide evidence of further relevant glyphosate sources. There is a potential that the sampling period was too short and that a small amount of the excreted glyphosate and AMPA could have been missed and the total excretion was underestimated. Although, these two minor issues have opposing shortcomings, which could cancel each other out.

The human elimination half-life of 9 h in our study is comparable to that of animal (rat) studies. In one rat study, a half-life of approximately 9.5 h was described after oral administration by gavage, 10.9 h in male rats and 8.1 h in female rats (FAO/WHO, 2017). Anadón et al. (2009) reported a half-life of 10.0 h after intravenous and 14.4 h after oral administration in rats. Connolly et al. (2019) found a half-life of 7.25 h in humans by calculating with urinary excretion rates and a range of 5.5 h up to 10 h when considering all calculation models. Faniband et al. (2017) reported first-order kinetics and a two-phase excretion with a half-life range for the rapid phase of 4–17 h. Overall, the published elimination half-lives have a good concordance.

4.2. AMPA

There is even less data available for AMPA than glyphosate. In our study, a low AMPA level was identified in the test meal consumed by the participants and therefore, they ingested a small amount of 1–2 µg of AMPA. If the amount of glyphosate in the test meal is assumed to be 100%, the fraction of AMPA concentration was only 0.85% (1.29% on a molar basis). After intravenous administration of glyphosate in rats, no transformation to AMPA was reported (EFSA, 2014; FAO/WHO, 2017; WHO, 1994). After oral administration in rats, glyphosate is mostly excreted unmetabolised with only small levels of AMPA observed (EFSA, 2014). Only 0.2–0.4% of a glyphosate dose of 10 mg kg^{-1} bw were found in the excreta as AMPA (FAO/WHO, 2017; WHO, 1997). In a study of bile duct cannulated rats, who received an oral dose of glyphosate of 10 mg kg^{-1} bw, only 0.07–0.19% of the dose was excreted in urine as AMPA (EFSA, 2014). In one rat study (Anadón et al., 2009), a relatively high transformation rate of glyphosate to AMPA concentration of 6.49% was found (9.88% on a molar basis). The data was measured in plasma and calculated by comparing the area under the concentration-time curves. Data on urinary excretion was not reported in this study. A small amount of human data is available from poisoning cases with herbicides. In seven cases of acute human intoxication, the ratio of urine concentrations of glyphosate/AMPA ranged from 243 to 7863 (Zouaoui et al., 2013). Only one study was available concerning the kinetics of AMPA alone. Wistar rats were fasted for 4 h and then given a single oral dose of 6.7 mg kg^{-1} bw AMPA by gavage. Overall, 20% of the dose was excreted in their urine within 120 h (18% within 24 h) (WHO, 1997).

Considering this data, the following two options are probable:

Option 1: There is a certain transformation rate of glyphosate to AMPA and all or most of the AMPA excreted in urine is a glyphosate metabolite. If all urinary AMPA stems from metabolised glyphosate then approximately 0.2% of the glyphosate dose on a concentration basis is excreted in urine as AMPA (see Fig. 1b, y axis on the right side). This corresponds to approximately 0.3% of the dose on a molar basis.

Option 2: There is no transformation to AMPA and the AMPA excreted in urine originated exclusively from the ingested AMPA. If this was the case, then AMPA is absorbed and excreted to a much higher degree than glyphosate. If no metabolism is assumed, approximately 23% of the AMPA dose is excreted in urine (see Fig. 1b, y axis on the left side). It is improbable that AMPA is so much more highly absorbed than glyphosate.

It is more probable that it is a combination of both options.

In this study, it was not possible to deduce where the urinary excreted AMPA originated. It could have been a metabolite of glyphosate and/or unchanged absorbed AMPA. Further studies administering pure glyphosate alone and pure AMPA alone are necessary.

The AMPA concentration in urine is often only slighty lower than that of glyphosate (Niemann et al., 2015). To date, there was no convincing explanation for this observation as it is generally agreed that glyphosate residues are higher than those of AMPA in a European diet. The intake of AMPA from other sources, e.g. as a breakdown product of certain detergents such as (nitrilotris(methylene))triphosphonic acid, has been postulated. Although this is the most plausible source, no studies to date have demonstrated this as a source or route of human exposure. The result of our experiment can contribute to partially explain these observations, either because of the higher than expected transformation rate of glyphosate to AMPA or the relatively better AMPA absorption.

4.3. Estimating glyphosate intake via urine sample measurement

Our data suggest that the glyphosate concentration in relation to the creatinine concentration and the glyphosate excretion rate are equally good markers for glyphosate intake (see Fig. 4a and b). To calculate the excretion rate, the sampling duration and urine volume must be known. Therefore, to interpret spot urine samples, it would be easier to use the glyphosate concentration in relation to the creatinine concentration. Of course, in both cases, it is also important to know the time interval since the suspected exposure.

5. Conclusion

In humans, the proportion of urinary excretion of glyphosate is only approximately 1% and therefore much lower than previously assumed based on animal data. Consequently, the glyphosate intake could be approximately 20 times higher than previously assumed when using urinary data. However, human systemic availability is most likely also 20 times lower than in rats, which gives the risk assessment a certain margin of conservatism.

Further human toxicokinetic studies should be performed individually. In these studies, the determination of blood concentrations is necessary to improve human bioavailability data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijheh.2020.113526.

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